

AES-128 Hardware Encryption Accelerator

Experiment 1 - Beginner | V-SPACE FPGA Build Challenge 2026

Team	Beyond Boolean
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Platform	PYNQ-Z2 xc7z020clg400-1
Recorded project tool	Vivado 2025.1.1

1. Objective

Implement a 128-bit AES encryption datapath in FPGA programmable logic and demonstrate its behavior using a known-answer test vector. The design uses an iterative round datapath and an on-chip Virtual Input/Output (VIO) core to enter the key/plaintext and observe results over JTAG.

2. Block Diagram

AES-128 hardware architecture

PYNQ-Z2 | iterative AES core with Vivado VIO control

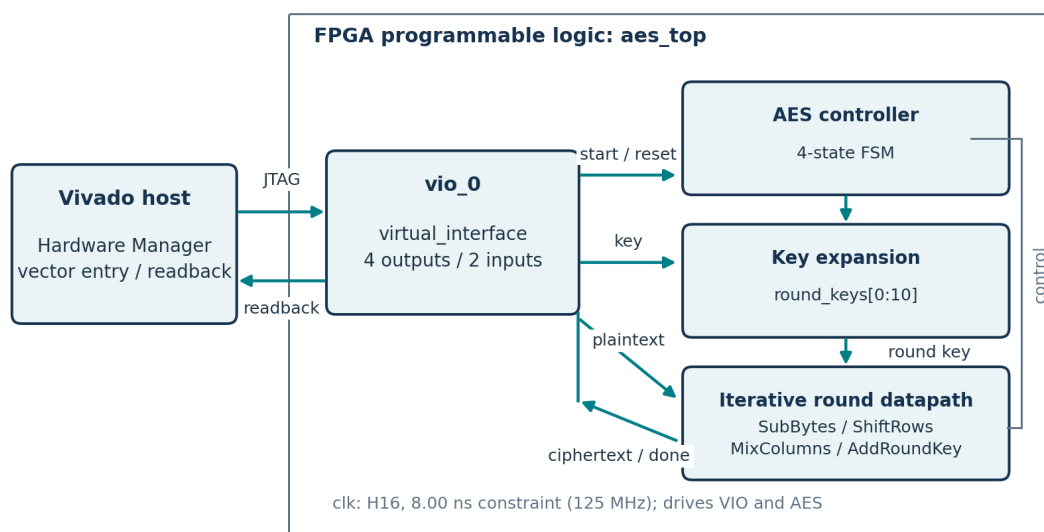


Diagram prepared from the recovered aes_top.v and aes_core.v.

Final AES round bypasses MixColumns. No AXI/PS overlay is present in this archived design.

Architecture drawn from the recovered RTL. The saved top is a pure RTL/VIO design; it contains no AXI DMA or processor-system block design.

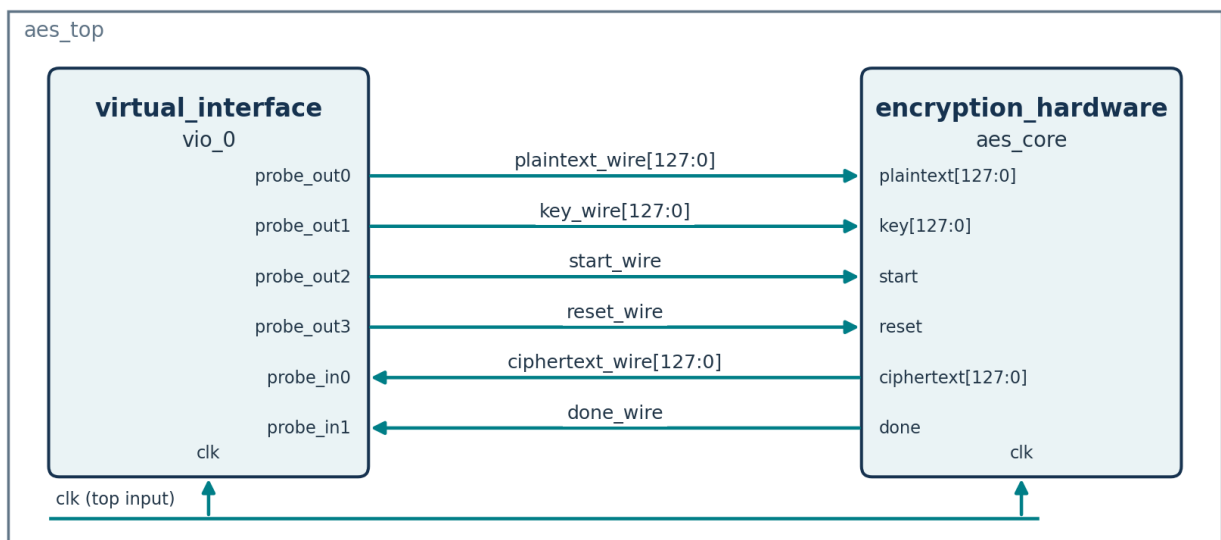
3. RTL Design

The design hierarchy contains `aes_top`, the AES submodule `aes_core` and the configured AMD VIO IP `vio_0`. The only external top-level RTL input is `clk`. The original source and IP configuration files are included in RTL; the full generated IP and project dependencies are preserved in the native source archive.

File	Role
<code>aes_top.v</code>	Instantiates <code>virtual_interface</code> (<code>vio_0</code>) and encryption hardware (<code>aes_core</code>); wires the six VIO signals and common clock.
<code>aes_core.v</code>	AES S-box, round constants, key expansion, SubBytes, ShiftRows, MixColumns, AddRoundKey and control state machine.
<code>vio_0.xci</code>	VIO configuration: four output probes and two input probes.
<code>pynq.xdc</code>	Clock input at H16, LVCMOS33; 8.00 ns clock constraint.

Structural RTL schematic: `aes_top`

Exact module instances, port directions and internal wire names from `aes_top.v`



Source-derived structural diagram; this is not a Vivado elaborated or synthesized netlist capture.

All six data/control connections and the common clock are shown. Bus widths: plaintext/key/ciphertext = 128.

Source-derived structural schematic of `aes_top.v`. This diagram documents the RTL wiring; it is not a Vivado netlist screenshot.

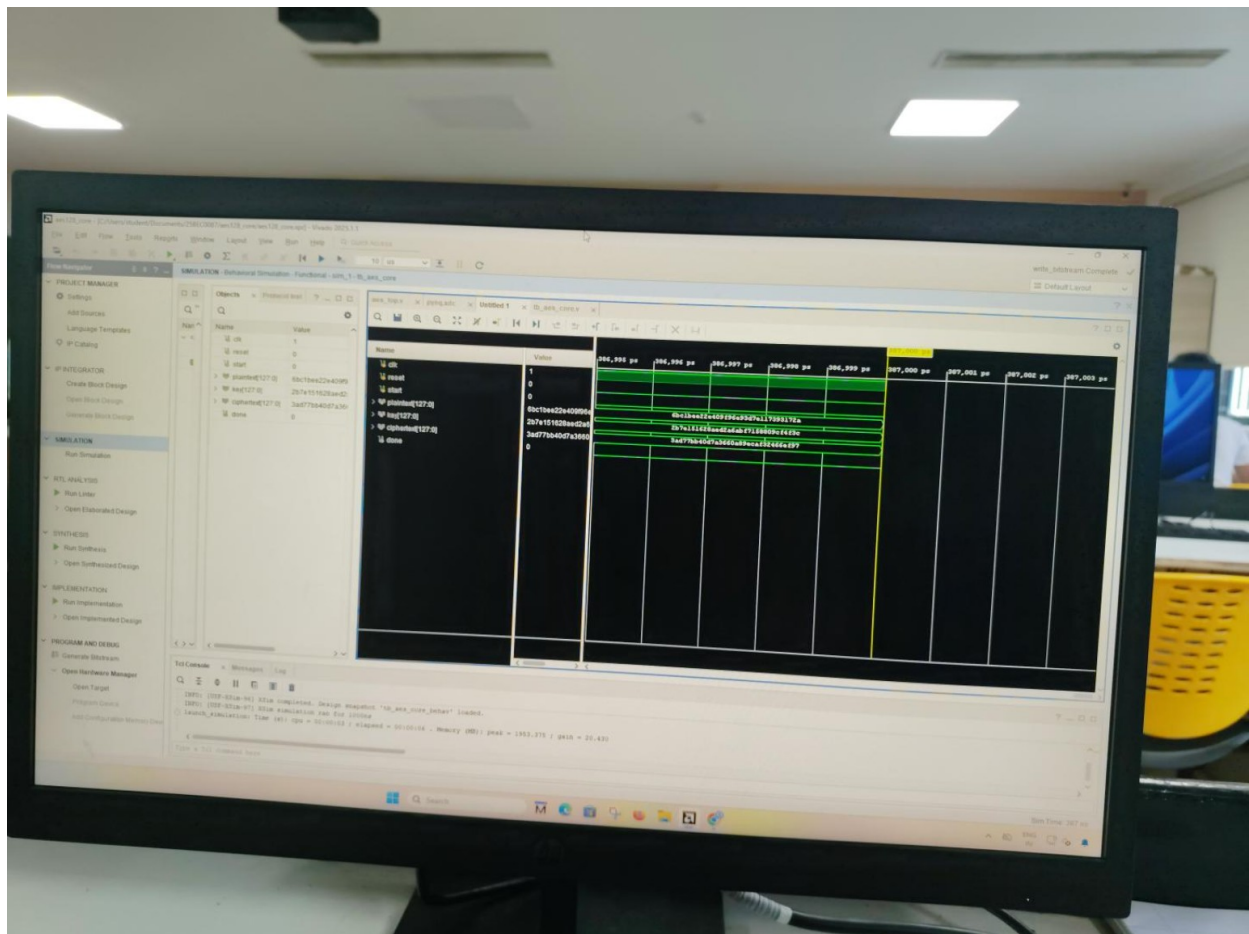
FSM state	Operation
<code>S_IDLE</code>	Clear done; on start, store the initial key and plaintext XOR key.
<code>S_KEY_EXP</code>	Compute and store round keys 1 through 10 over successive clocks.
<code>S_ROUNDS</code>	Reuse the round datapath; omit MixColumns in round 10.
<code>S_FINISHED</code>	Register ciphertext, assert done, and return to <code>S_IDLE</code> .

`done` is a one-clock pulse in this RTL: `S_IDLE` clears it on the following clock. The ciphertext register remains available until reset or the next completed encryption.

4. Simulation Results

The preserved behavioral simulation uses tb_aes_core with a 10 ns clock period (100 MHz). The archived transcript reports a passing comparison for the following single known-answer vector:

Signal	Archived hexadecimal value
Key	2b7e151628aed2a6abf7158809cf4f3c
Plaintext	6bc1bee22e409f96e93d7e117393172a
Expected ciphertext	3ad77bb40d7a3660a89ecaf32466ef97
Observed ciphertext	3ad77bb40d7a3660a89ecaf32466ef97



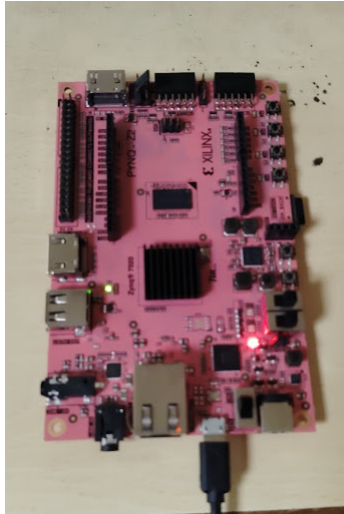
Original supplied simulation photograph. Its time window is tightly zoomed near the end of a historical run; it is not a newly rendered trace. The saved WDB is included for a clearer full-transaction capture.

Saved console status: >> **TEST PASSED** <<. The original transcript ends at 387 ns. Only one vector is preserved; this result does not establish a complete AES regression or implementation timing closure.

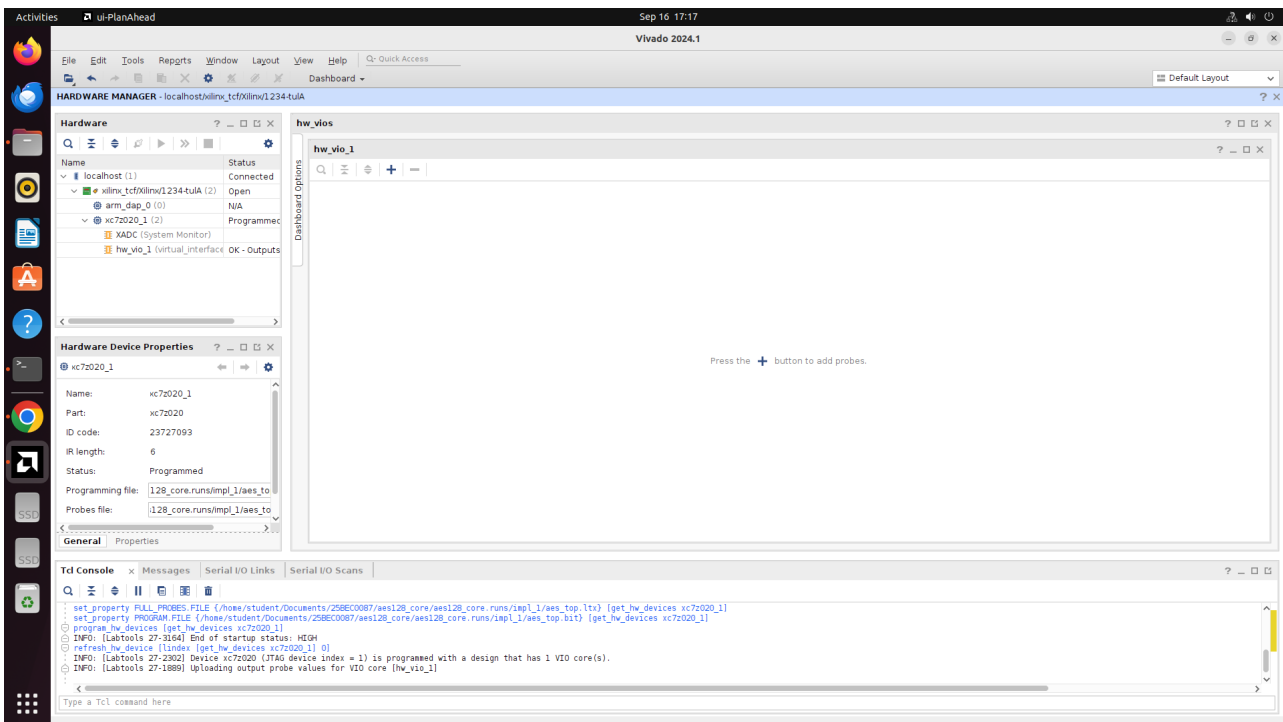
5. Hardware Implementation

The recovered files preserve synthesis/implementation reports, a routed checkpoint and the matching aes_top.bit and aes_top.ltx programming files. The supplied hardware screenshot shows the xc7z020 target programmed and the VIO core active. Physical-board photographs are also preserved.

Physical setup



Existing PYNQ-Z2 setup photo extracted from page 53 of the uploaded project conversation. Original resolution: 347 x 512 pixels.



Original 1920 x 1080 Hardware Manager screenshot: programmed FPGA and active VIO. The dashboard is not displaying a completed encryption in this image.

Hardware result evidence

The project conversation discusses a 21-second clip and reports that its beginning displayed ciphertext 3ad77bb40d7a3660a89ecaf32466ef97 before reset. The package preserves its 512 x 278 poster as hardware_output.jpg. The original clip is unavailable here, and the poster is too small to independently verify all digits. Replace this poster with a readable frame from the original recording and add the video URL before submission.

5. Hardware Implementation - Build Results

Metric	Archived integrated-design result
Target device	xc7z020clg400-1
Clock constraint	8.00 ns / 125 MHz
Slice LUTs	3,222 of 53,200 (6.06%)
Slice registers	3,912 of 106,400 (3.68%)
Setup WNS / TNS	-0.104 ns / -0.276 ns
Setup-failing endpoints	4
Hold WHS / failing endpoints	+0.028 ns / 0
Timing report verdict	Timing constraints are not met.
Routed DRC report	Five warning checks; no listed error/critical-warning checks.

Timing qualification

The saved routed build fails setup timing at the declared 125 MHz clock. The failing paths are in AES key-expansion logic. The generated bitstream and a reported functional demonstration do not remove this constraint violation. The behavioral simulation runs at 100 MHz and is a separate verification condition.

Resource accounting

The utilization figures above apply to the integrated AES, VIO and debug infrastructure. They should not be presented as isolated AES-core utilization. The original utilization, timing, DRC, methodology, route-status and other reports are included in Documentation/Implementation_Reports/.

Reproducibility

Extract FPGA_Project_Files/Native_Project/Project1_AES_Vivado_Source.zip, keep its aes128_core folder intact and open aes128_core.xpr in Vivado 2025.1.1. Original source and generated IP dependencies are preserved; disposable caches and run folders are excluded. The routed checkpoint, reports and waveform database are supplied separately. Hardware programming uses the BIT/LTX pair in FPGA_Project_Files/Programming/.

Future timing work

A newer build should optimize the failing key-expansion paths and validate operation with the actual board clock. Relaxing the XDC period alone does not change the physical clock. Retain this report qualification until a subsequent routed report demonstrates closure.

6. Applications

This AES primitive can form part of an embedded cryptographic coprocessor, secure-communication subsystem or FPGA teaching platform. Its VIO interface is useful for laboratory testing and demonstration. A production system would additionally require suitable buffering/interfaces, secure key management, an appropriate encryption mode and authentication.

7. Conclusion

The recovered project demonstrates an iterative AES-128 RTL implementation, a passing archived single-vector behavioral simulation, FPGA synthesis/implementation, generated programming files and recorded physical-board setup/programming activity. The engineering workflow reaches the bitstream and demonstration stages.

Remaining limitations are the negative setup slack in the preserved build, limited regression coverage, and incomplete final hardware/video evidence. The ciphertext register persists, while done is a pulse; hardware readback should interpret this handshake correctly. Future work includes timing closure, additional vectors and corner-case testing, and a documented runtime interface.

Evidence and submission status

Evidence type	Included / remaining
Original files	RTL, XDC, VIO configuration, testbench, native project, BIT/LTX, saved WDB and reports.
Simulation	Original passing transcript and historical waveform photo; separate simulation PDF included.
Authored figures	Block diagram and source-derived structural RTL schematic.
Physical evidence	Board photos and programmed-VIO screenshot; result poster requires a readable replacement.
Video	Video_Link.txt is present and blank; insert the actual finished demonstration-video URL.

Source record

1. Original aes128_core.xpr.zip: recovered RTL, testbench, simulate.log, saved WDB, routed reports and programming outputs.
2. documents req.pdf: required folder layout and seven report sections.
3. Uploaded project conversation PDF, pp.38-54: archive transfer, VIO programming, board photo and video-poster context.
4. Original supplied PNG images: waveform photograph, source hierarchy/editor, programmed-VIO dashboard and loaded vectors.

This report was assembled from the preserved project evidence. No new Vivado simulation, synthesis, implementation or physical-board test was performed during document preparation.